Rapid Application Development

Chapter 1: Introduction to RAD ✓

Chapter 2: Key Elements of RAD ✓

Chapter 3: Agile Software Development ✓

Chapter 4: Information Requirements Analysis ✓

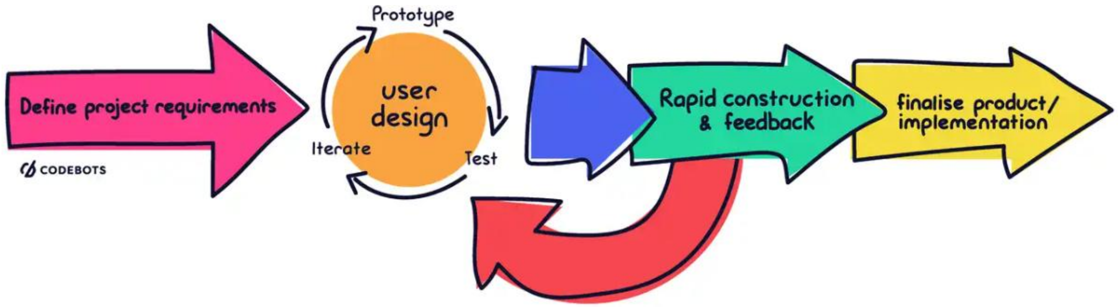
Chapter 5: Analysis Process ✓

Chapter 6: Testing ✓

Chapter 7: Quality Assurance and Implementation ✓

Chapter 8: Best Practices and Tools ✓

Chapter 9: Software Project Management



Chapter 9: Software Project Management

Work breakdown structure

Project cost estimation

- Break-even analysis
- Cash-flow analysis
- Present value analysis

Project scheduling using Gantt chart and PERT diagrams

Work breakdown structure

- Systems analysts are responsible for **completing projects on time** and **within budget** and for **including the features promised**.

- In order to accomplish all three of these goals, a project needs to be broken down into smaller tasks or activities.

- These tasks together make up a **Work Breakdown Structure (WBS).**

Work breakdown structure properties

- Each task or activity **contains one deliverable, or tangible outcome**, from the activity.
- Each task can be **assigned to a single individual or a single group**.
- Each task has a **responsible person monitoring and controlling performance**.

The main method for developing a WBS is **decomposition**, or starting with large ideas and then breaking them down into manageable activities.

There are different types of work breakdown structures.

- Product-oriented
- Process-oriented

Product-oriented work breakdown structures

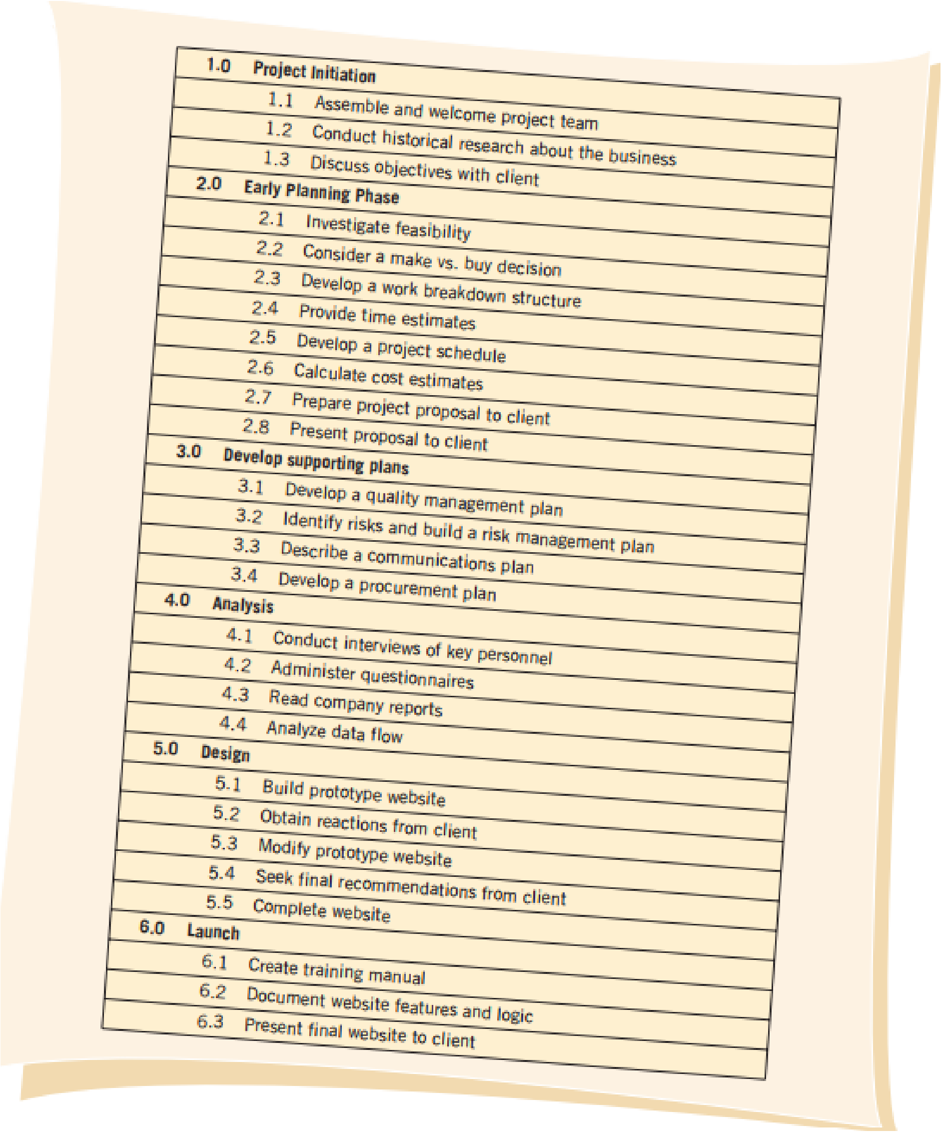
Example:

- Building a website can be broken down into many parts, with each set of pages having a **specific purpose**.
- You could divide a website into its home page, product description pages, a FAQ page, a contact page, and an ecommerce page.
- Each of these pages could contain activities that you could use in your work breakdown structure.

Process-oriented work breakdown structures

Example:

- This is shown in the figure.
- This type of WBS is typical in systems analysis and design.
- In this example, we show the **development of a website**, but rather than show the development of each page.
- This example emphasizes the importance of **each phase in the systems development life cycle.**



1.0	**Project Initiation**
1.1	Assemble and welcome project team
1.2	Conduct historical research about the business
1.3	Discuss objectives with client
2.0	**Early Planning Phase**
2.1	Investigate feasibility
2.2	Consider a make vs. buy decision
2.3	Develop a work breakdown structure
2.4	Provide time estimates
2.5	Develop a project schedule
2.6	Calculate cost estimates
2.7	Prepare project proposal to client
2.8	Present proposal to client
3.0	**Develop supporting plans**
3.1	Develop a quality management plan
3.2	Identify risks and build a risk management plan
3.3	Describe a communications plan
3.4	Develop a procurement plan
4.0	**Analysis**
4.1	Conduct interviews of key personnel
4.2	Administer questionnaires
4.3	Read company reports
4.4	Analyze data flow
5.0	**Design**
5.1	Build prototype website
5.2	Obtain reactions from client
5.3	Modify prototype website
5.4	Seek final recommendations from client
5.5	Complete website
6.0	**Launch**
6.1	Create training manual
6.2	Document website features and logic
6.3	Present final website to client

Project cost estimation

Identifying Benefits and Costs

- Tangible benefits are **advantages** measurable through the use of the information system.
- Intangible benefits are difficult to measure
- Tangible costs are **costs** that a systems analyst and the business's accounting personnel can accurately project.
- Intangible costs are difficult to estimate and may not be known
- Both tangible and intangible costs must be taken into account when systems are considered.

Tangible benefits	Intangible benefits
• Increase in the speed of processing • Gain entry to information that would otherwise be unreachable • Access to information on a more timely basis • The advantage of the computer’s superior calculating power • Decreases in the amount of employee time needed to complete specific tasks	• Improving the decision-making process • Enhancing accuracy • Becoming more competitive in customer service • Maintaining a good business image • Increasing job satisfaction
Tangible costs	**Intangible costs**
• Cost of equipment • Cost of resources • Cost of systems analysts’ time • Cost of programmers’ time • Employees’ salaries	• Losing a competitive edge • Losing the reputation of being first • Declining company image • Ineffective decision making

Comparing Costs and Benefits

There are many well-known techniques for comparing the costs and benefits of the proposed system.

- Break-even analysis
- Cash-flow analysis
- Present value analysis

Break-even analysis

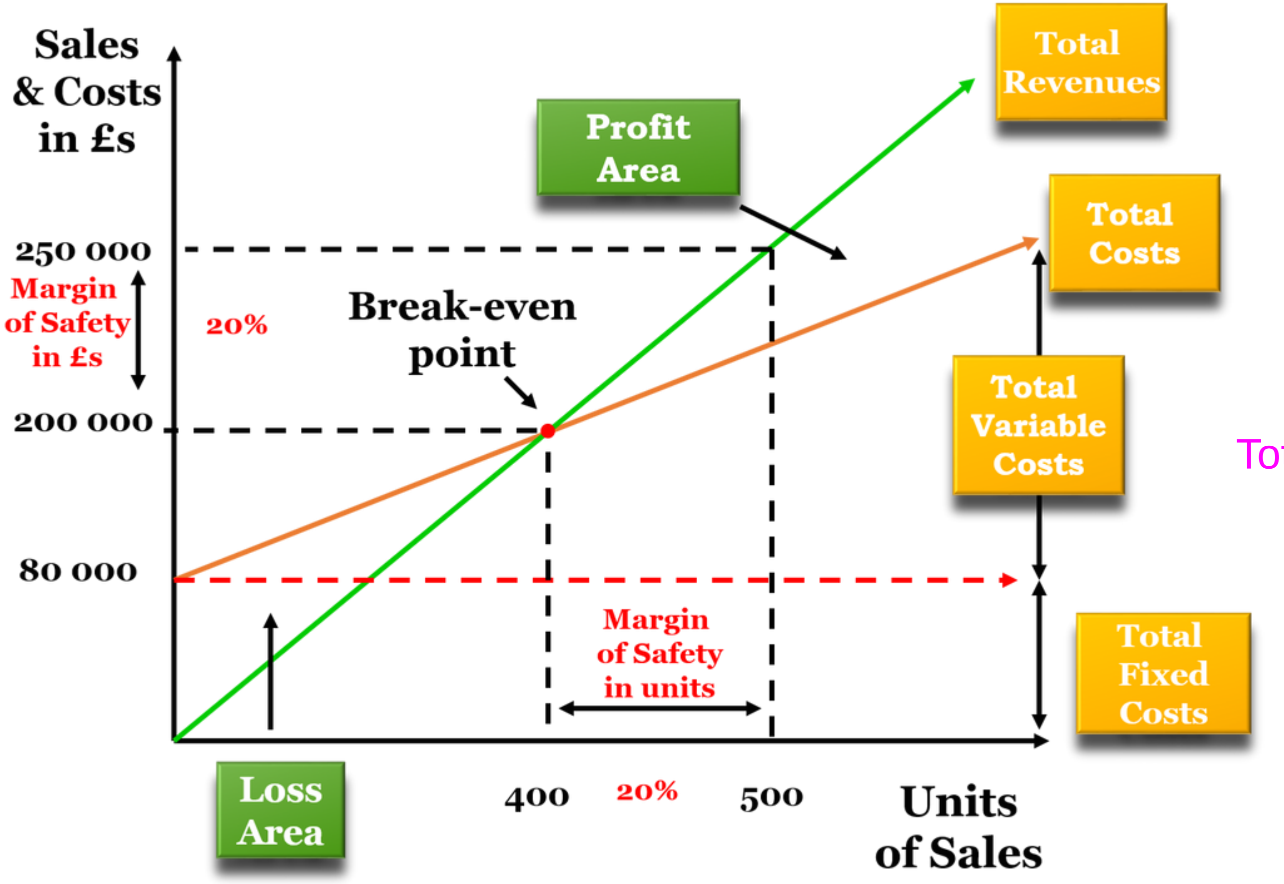
- **A break-even analysis** helps company owners find the point at which their total costs and total revenue are equal, also known as the Break-Even Point (BEP).

- **A Break-even chart** is the graphical representation of the relation between cost and revenue at a given time.

Construction of break-even chart

- **Revenue -** Revenue is the total money generated by selling goods within a specific time, before dedicated any expenses.
- **Fixed cost -** Fixed cost does not change with an increase or decrease in the number of goods or services sold.
- **Variable Cost -** Variable costs are a corporate expense that changes in proportion to production output.

Construction of break-even chart

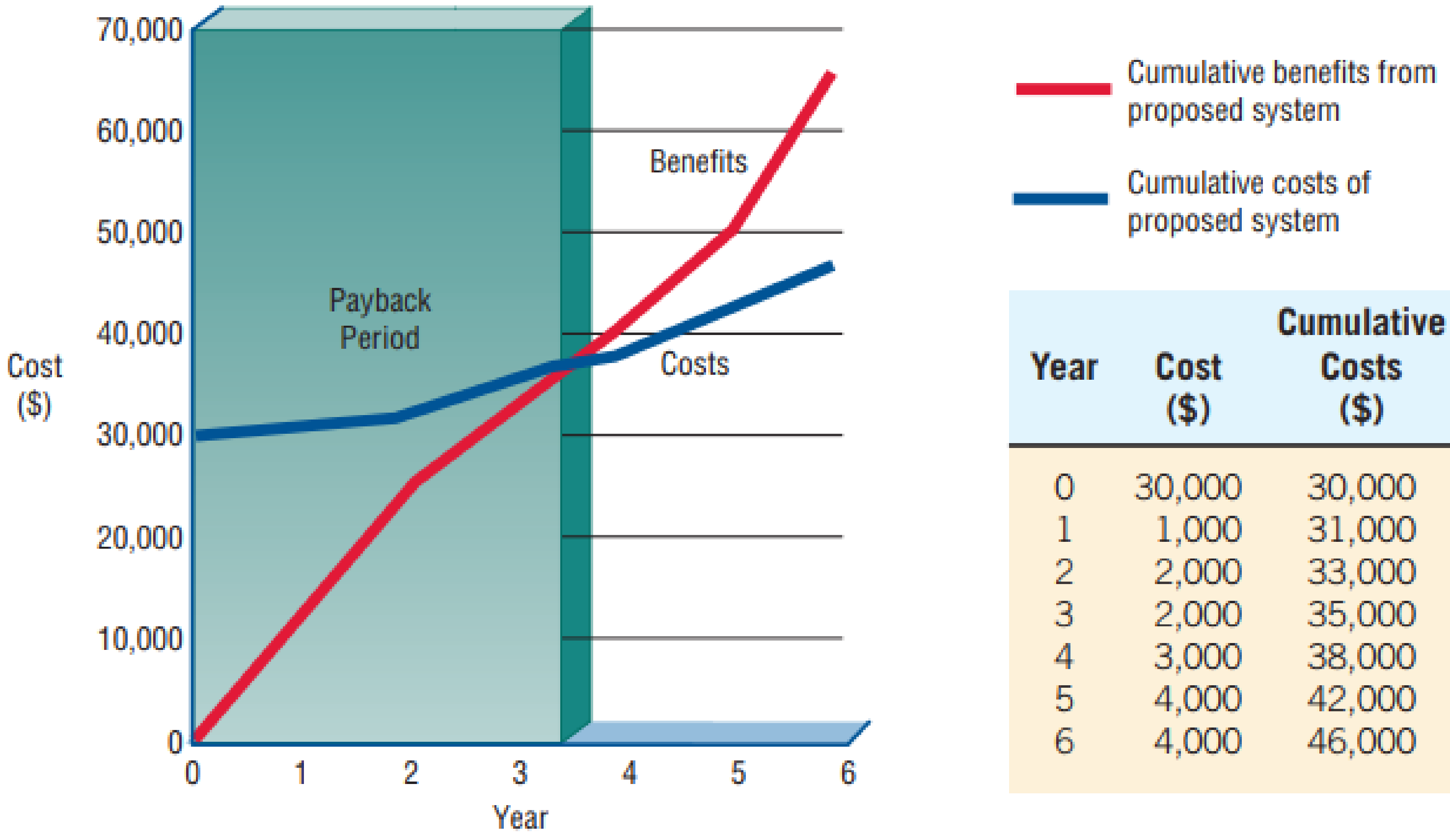


Total costs = Fixed cost + Variable cost

Margin of Safety = ((Actual Sales – Break-even Sales)/ Actual Sales)*100%

The higher margin of safety, the safer project.

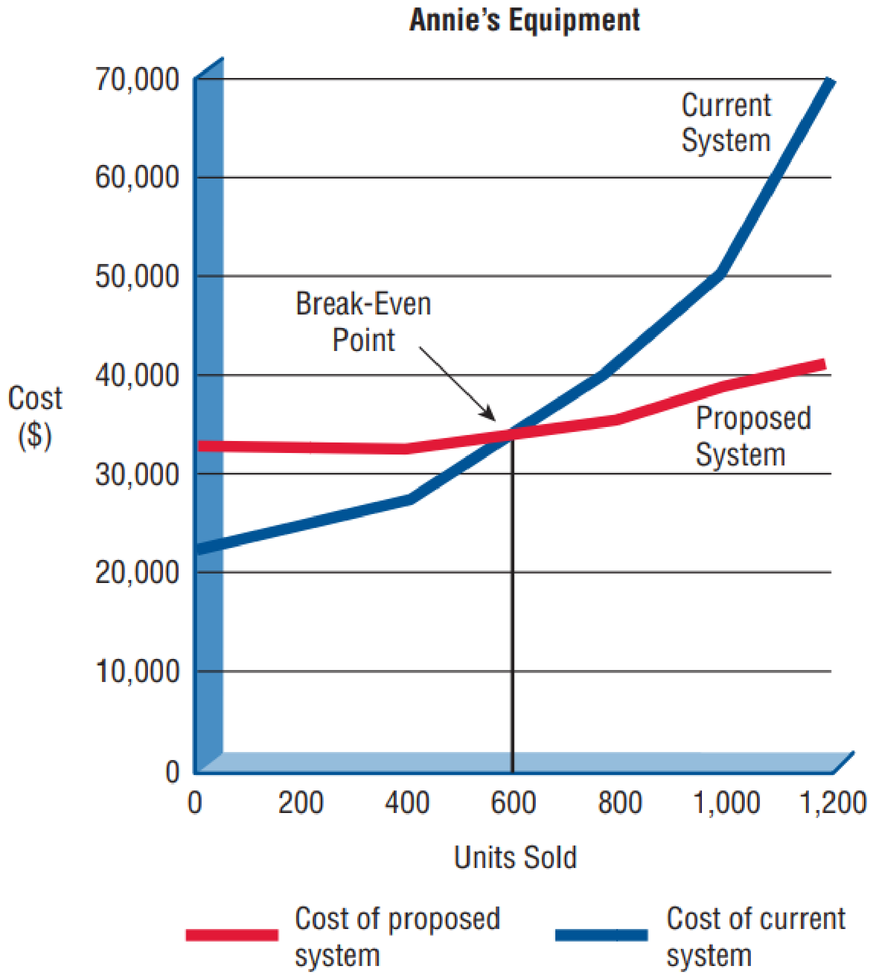
- Break-even analysis can also determine how long it will take for the benefits of the system to **pay back the costs of developing it.**



Year	Cost ($)	Cumulative Costs ($)	Benefits ($)	Cumulative Benefits ($)
0	30,000	30,000	0	0
1	1,000	31,000	12,000	12,000
2	2,000	33,000	12,000	24,000
3	2,000	35,000	8,000	32,000
4	3,000	38,000	8,000	40,000
5	4,000	42,000	10,000	50,000
6	4,000	46,000	15,000	65,000

- Payback period – nearly three and a half years

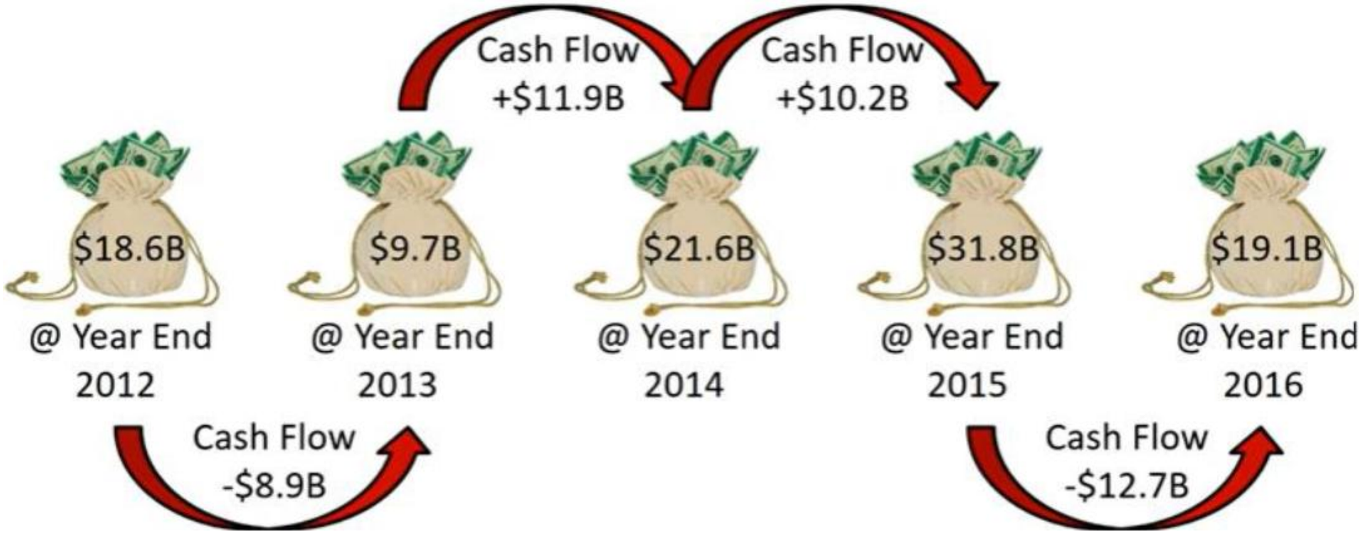
Break-even analysis for a proposed inventory system.



- A small store maintains inventory using a **manual system.** As volume rises, the **costs of the manual system rise at an increasing rate**.
- A new **computer system** would cost a substantial sum up front, but the **incremental costs for higher volume would be rather small.**
- The graph shows that the computer system would be cost-effective if the business sold about 600 units per week.
- **BEP** - The point at which the total cost of the current system and the proposed system intersect.

Cash-flow analysis

- The cash flow from **investing activities** looks at **long-term uses of cash.**
- This may include **the sale or purchase** of a fixed asset like property or equipment.
- This involves examining the cash inflows and outflows from various activities during a specific period, including operations, investments, and financing.



- Examines the **direction, size**, and **pattern of cash flow** that is associated with the proposed information system.
- Cash-flow analysis is used to **determine when a company will begin to make profit**.

	Year 1 Quarter 1	Quarter 2	Quarter 3	Quarter 4	Year 2 Quarter 1
Revenue	$5,000	$20,000	$24,960	$31,270	$39,020
Costs					
Software development	10,000	5,000			
Personnel	8,000	8,400	8,800	9,260	9,700
Training	3,000	6,000			
Equipment lease	4,000	4,000	4,000	4,000	4,000
Supplies	1,000	2,000	2,370	2,990	3,730
Maintenance	0	2,000	2,200	2,420	2,660
Total Costs	26,000	27,400	17,370	18,670	20,090
Cash Flow	–21,000	–7,400	7,590	12,600	18,930
Cumulative Cash Flow	–21,000	–28,400	–20,810	–8,210	10,720

Present value analysis

- Present value analysis assists a **systems analyst** in illustrating to **business decision-makers** the time value of an investment in the information system and its associated cash flow.

- Present value is a way to **assess** all the economic **expenses and revenues** of the information system **over its economic life** and to compare costs today with future costs and today's benefits with future benefits.

Without considering present value

FV

	Year						
	1	2	3	4	5	6	Total
Costs	$40,000	42,000	44,100	46,300	48,600	51,000	272,000
Benefits	$25,000	31,200	39,000	48,700	60,800	76,000	280,700

The benefits appear to outweigh the costs.

- System costs total = $272,000 over six years, and benefits total = $280,700.
- Therefore, we might conclude that the **benefits outweigh the costs.**
- **Benefits only started to surpass costs** in the fourth year, however, dollars in the sixth year will not be equivalent to dollars in the first year.
- Example: a $1 investment at 7 percent today will be worth $1.07 at the end of the year and will double in approximately 10 years.

Present value and Future value

Example of future value:

How much money will I have after one year, if I invest **$100** at an expected **20%** annual return?

$$\$100 \xrightarrow{\text{x } 1.2} \$120$$

Example of present value:

How much money do I invest today to achieve **$120** after one year at an expected **20%** annual return?

$$\$100 \xleftarrow{/\ 1.2} \$120$$

$$PV = \frac{FV}{(1 + r)^n}$$

PV = Present value	r = Discount rate / Rate of return
FV = Future value	n = Number of periods

PV = $100

FV = $144

20% annual return
r = 0.2

n = 2

Future Value Formula

$$\$100 \xrightarrow{X\ 1.2} \$120 \xrightarrow{X\ 1.2} \$144$$

$\$100 \text{ X } (1.2)^2 = \$144$

$\boxed{PV \text{ X } (1+r)^n = FV}$

Present Value Formula

$$\$100 \xleftarrow{/\ 1.2} \$120 \xleftarrow{/\ 1.2} \$144$$

$\$100 = \$144 / (1.2)^2$

$$\boxed{PV = \frac{FV}{(1+r)^n}}$$

Now **Future**

2024 **2029**

Present Value **Future Value**

Earlier **Now**

2020 **2024**

Present Value **Future Value**

Taking into account present value → In calculating the multipliers in this table, the discount rate, *r*, is assumed to be 0.12

Calculating Now
FV

Calculating Earlier
PV

	Year						
	1	**2**	**3**	**4**	**5**	**6**	**Total**
Costs	\$40,000	42,000	44,100	46,300	48,600	51,000	
Multiplier	.89	.80	.71	.64	.57	.51	
Present Value of Costs	35,600	33,600	31,311	29,632	27,702	26,010	183,855
Benefits	\$25,000	31,200	39,000	48,700	60,800	76,000	
Multiplier	.89	.80	.71	.64	.57	.51	
Present Value of Benefits	22,250	24,960	27,690	31,168	34,656	38,760	179,484

The conclusion is that the costs are greater than the benefits

- Present values of both costs and benefits are then calculated using these multipliers. → $\frac{1}{(1+r)^n}$
- When this step is done, the total benefits (measured in today's dollars) are \$179,484 and are thus less than the costs (also measured in today's dollars) \$183,855.
- The conclusion to be drawn is that the proposed system is not worthwhile if present value is considered.

Guidelines for analysis

- Use break-even analysis if the project needs to be **justified in terms of cost**, not benefits, or if **benefits do not substantially improve with the proposed system.**

- Use cash-flow analysis when the **project is expensive relative to the size of the company** or when the business would be **significantly affected by a large drain** (even if temporary) on funds.

- Use present value analysis when the **payback period is long or when the cost of borrowing money is high.**

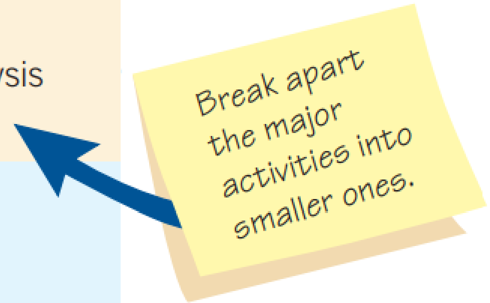
Project scheduling using Gantt chart and PERT diagrams

Project planning includes all the activities required to **select a systems analysis team**, **assign members of the team** to appropriate projects, **estimate the time** required to complete each task, and **schedule the project** so that tasks are completed in a timely fashion.

FIGURE 3.16

Beginning to plan a project by breaking it into three major activities.

Phase	Activity
Analysis	Data gathering Data flow and decision analysis Proposal preparation
Design	Data entry design Input design Output design Data organization
Implementation	Implementation Evaluation



- The **most difficult part** of project planning is the crucial step of **estimating the time** it takes to complete each task or activity.

Activity	Detailed Activity	Weeks Required
Data gathering	Conduct interviews	3
	Administer questionnaires	4
	Read company reports	4
	Introduce prototype	5
	Observe reactions to prototype	3
Data flow and decision analysis	Analyze data flow	8
Proposal preparation	Perform cost-benefit analysis	3
	Prepare proposal	2
	Present proposal	2

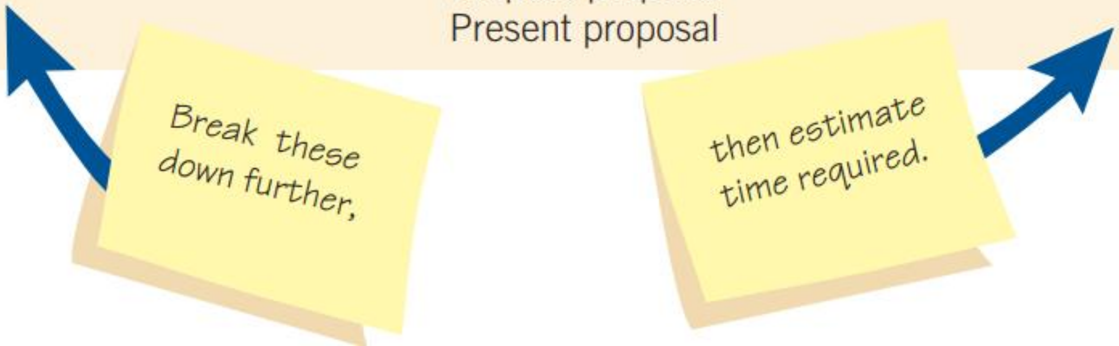


FIGURE 3.17

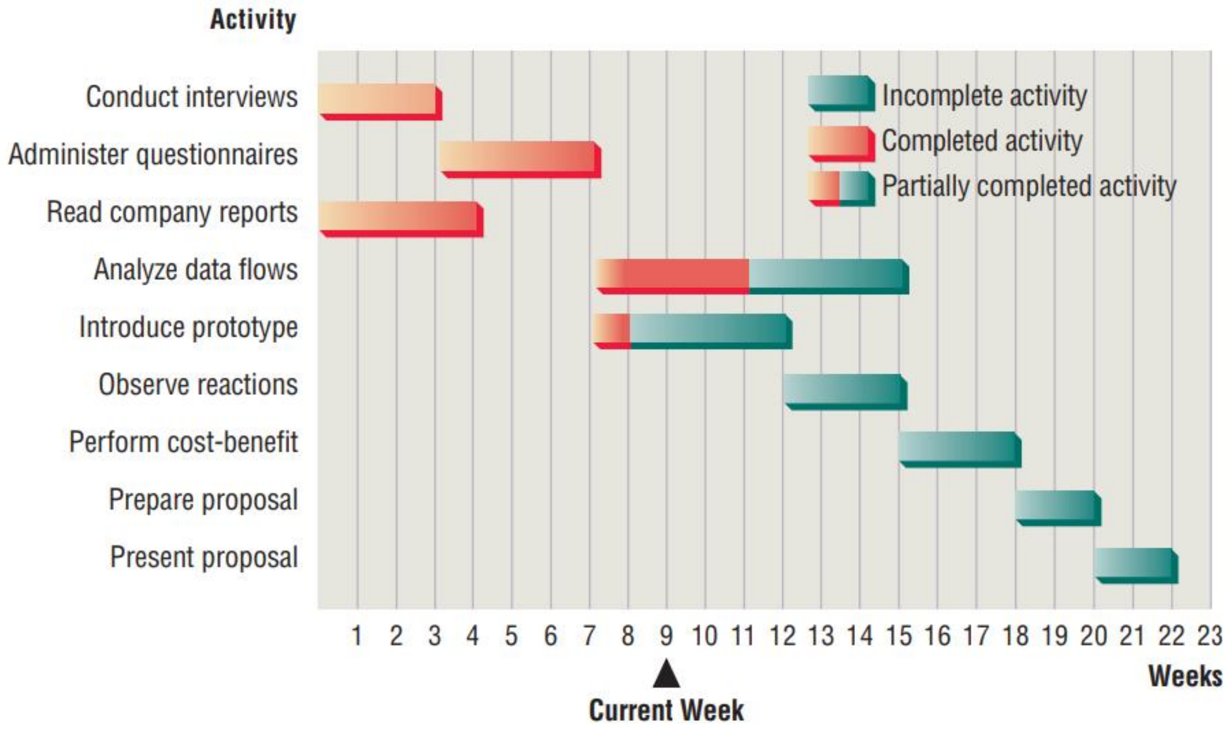
Refining the planning and scheduling of analysis activities by adding detailed tasks and establishing the time required to complete the tasks.

Gantt charts for project scheduling

- A Gantt chart is a tool that enables you to easily schedule tasks.
- It is a chart on which bars represent tasks or activities.
- The length of each bar represents the relative length of the task.

FIGURE 3.18

Using a two-dimensional Gantt chart for planning activities that can be accomplished in parallel.

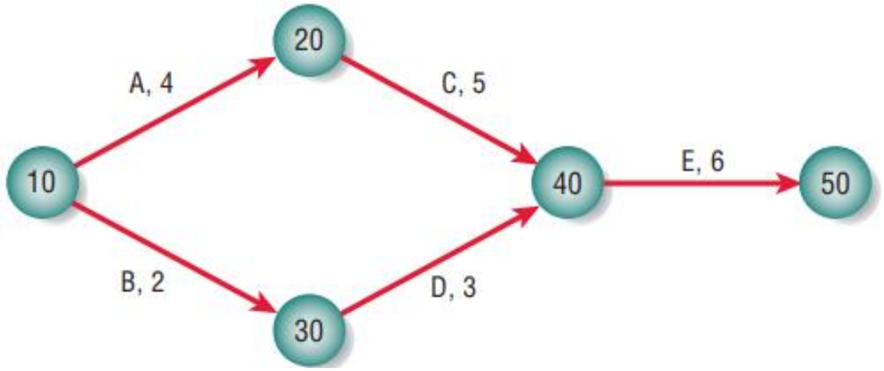


Advantages

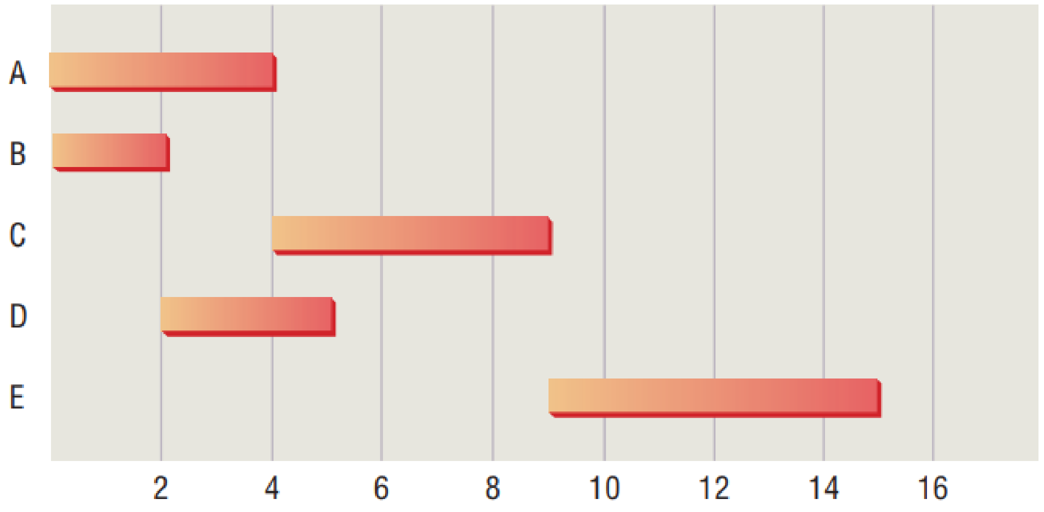
- The main advantage of a Gantt chart is its **simplicity.**
- Not only easy to use, but also **lends itself to worthwhile communication with end users.**
- The bars representing **activities or tasks are drawn to scale**; that is, the **size of the bars indicates the relative length of time** it will take to complete each task.

PERT diagrams for project scheduling

- PERT is an acronym for Program Evaluation and Review Technique.
- Represented by a network of **nodes and arrows**.
- PERT is useful when activities can be done in **parallel** rather than in sequence.
- **Activities expressed as arrows** in the PERT diagram, represent bars in the Gantt chart.
- **No direct relationship** with the length of the arrows and activity durations.
- **Circles are called milestones** and identified by numbers, letters, or any other arbitrary form of designation.
- **Precedence** - Activities need to be completed before new activities may be undertaken. (**A** need to be completed before **C** started)



Gantt charts vs. PERT



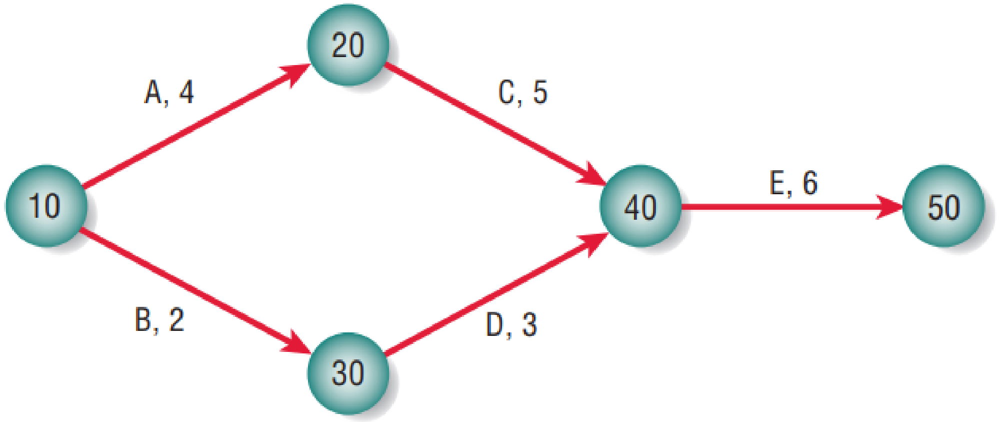
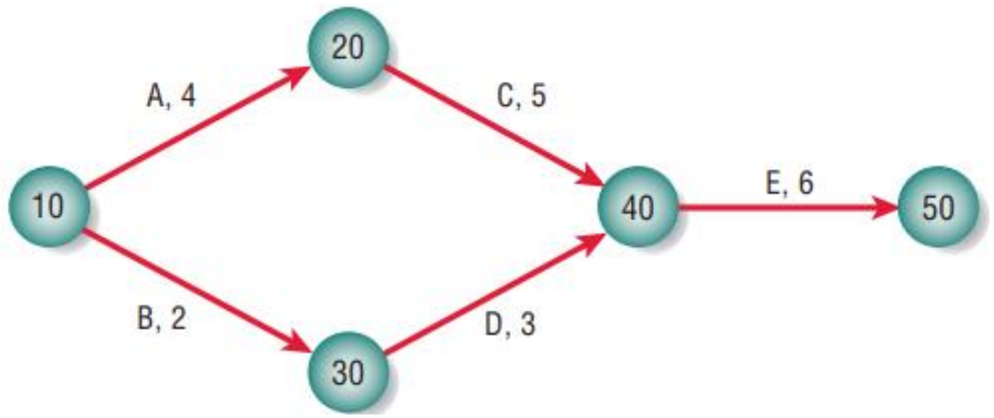


FIGURE 3.19

A Gantt chart compared with a PERT diagram for scheduling activities.

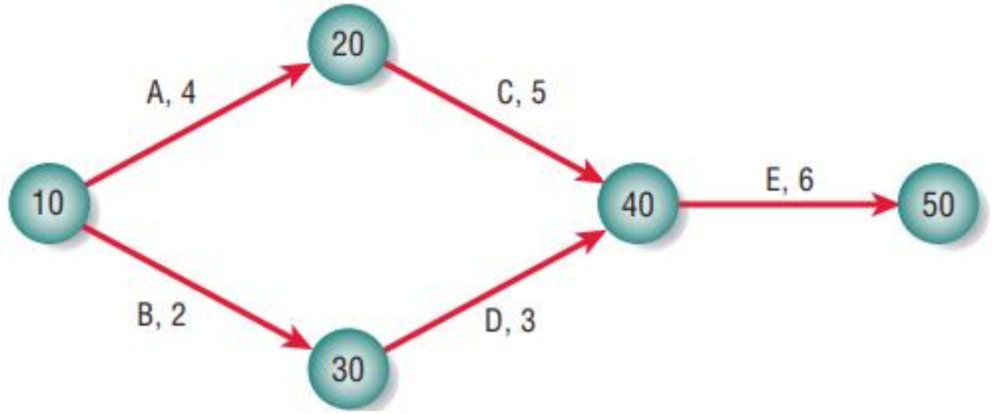
- In reality activity C may not be started until activity A is completed.
- **Precedence is not indicated at all in the Gantt chart**, so it is not possible to tell whether activity C is scheduled to start on day 4 on purpose or by coincidence.

PERT – Critical Path



- The **longest path** is referred to as the critical path.
- **Critical** path is the path that will cause the **whole project to fall behind schedule if one day's delay** is encountered on it.
- E.g.: If you are delayed one day on path 10–20–40–50, the entire project will take longer.

PERT – Slack time

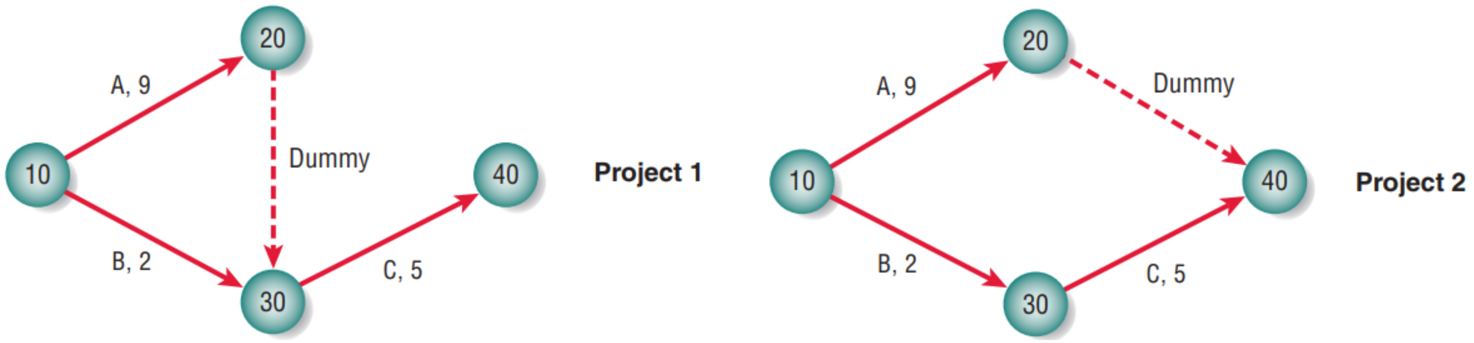


- A slide delay in **non-critical path** is called slack time.
- E.g.: If you are delayed one day on path 10–30–40–50, the entire project **will not suffer**.
- Leeway (allowable time lost) on noncritical paths – **slack time.**

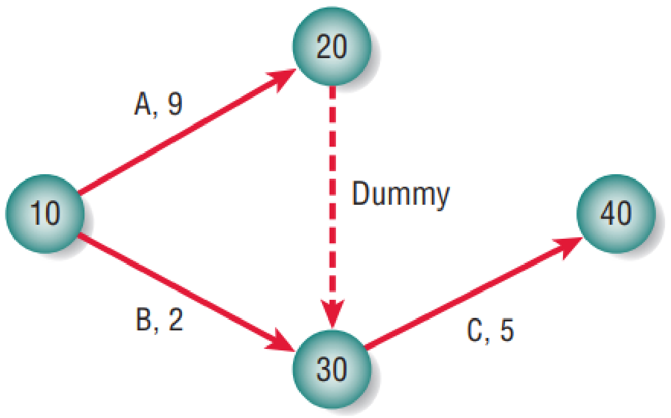
PERT – Dummy paths

FIGURE 3.20

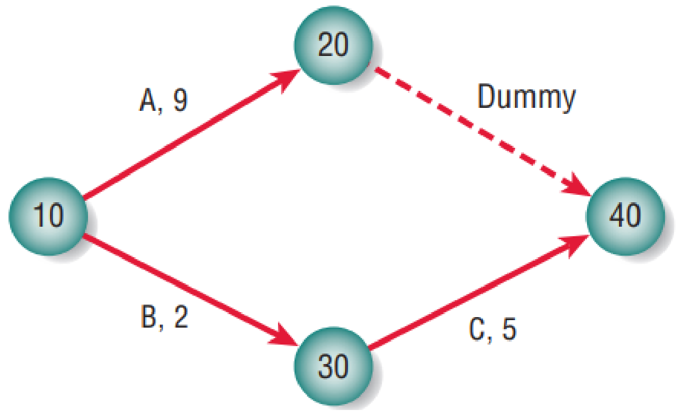
Precedence of activities is important in determining the length of a project when using a PERT diagram.



- Dummy paths represented by a **dashed line** in the PERT diagram.
- This dummy activity is pretended in the sense that it requires **no time or resources**.
- It enables us to draw a PERT representation that **correctly maintains the appropriate precedence** relationships.
- Project 1 and project 2 are quite different, and the **way the dummy is drawn makes the difference clear.**



Project 1



Project 2

- In project 1, **activity C can be started only if both A and B are finished** because all arrows coming into a node must be completed before leaving the node.
- In project 2, however, **activity C requires only activity B's completion** and can therefore **be under way while activity A is still taking place**.
- Project 1 takes 14 days to complete, whereas project 2 takes only 9 days.

Advantages

There are many reasons for using a PERT diagram over a Gantt chart. The PERT diagram allows:

- Easy identification of the order of precedence
- Easy identification of the critical path and thus critical activities
- Easy determination of slack time

Q & A

Q1. List two examples for tangible benefits and intangible cost for your project.

Q2. State two advantages of Gantt charts over PERT diagrams.

Studied so far

Software Project Management

- Work breakdown structure
  - ✓ Work breakdown structure properties
- Project cost estimation
  - ✓ Comparing Costs and Benefits, Break-even analysis, Cash-flow analysis, Present value analysis
- Project scheduling using Gantt chart and PERT diagrams
  - ✓ Gantt charts for project scheduling, PERT diagrams for project scheduling

Outcomes

Intended Learning Outcomes (ILOs)	Achieved
Describe the concepts of software development methodologies	100%
Demonstrate the importance of Rapid Application Development (RAD) and its key elements	100%
Discuss how systems analysts interact with users, management, and other information systems professionals for gathering requirements	100%
Analyse the development lifecycle of a given software project	100%
Develop a software rapidly by best practices and tools	100%